

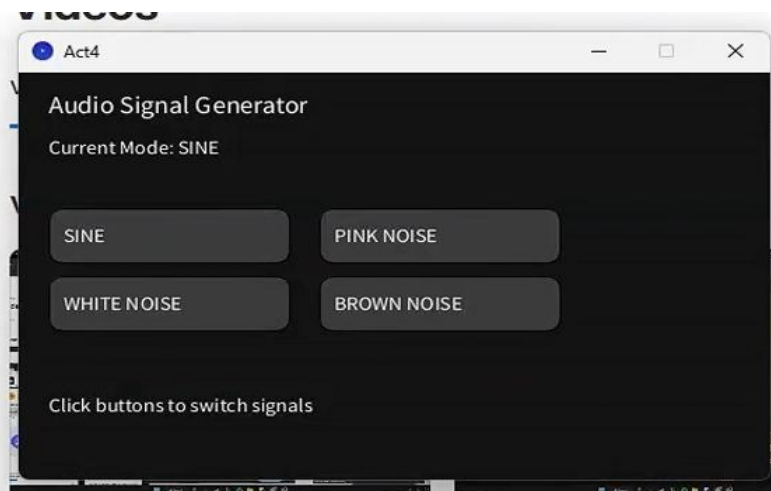


Digital Signal Processing
Engr. Jobert DC. Cadiz

Finals Period Act	Activity#4: Real-Time Audio Signal Processing with Interactive Buttons in Processing Members: Ma. Kristal Czassandra A. Rivera Anwil Shien Oli Harold James Enciso John Jefferson Catibog Kriztoffer Ryan Hernandez Joshua Nuevo Lanze Ruzzel Orleans Section: BSCPE Ladderize Group Number:	Grade
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I - Objectives:

Students will generate and modify real-time audio signals (noise or sine waves) using Processing software. The interactive buttons in the Processing interface will allow students to switch between different signals (e.g., white noise, pink noise, brown noise, or a sine wave). Sample output:



There are only 10 types of people in the world: those who understand binary and those who don't.

II – Tools/Materials:

Processing Software

- Python 3.x idle or any preferred ide (vs code, pycharm, etc.)
- Pygame library (installable via pip install pygame)
- Numpy library (installable via pip install numpy)
- Computer with audio output capability (speakers/headphones)

III – Program Codes

Note: Code should have proper comments: It should include group number, members and explanation of the program code.

"""

CPE017 (Digital Signal Processing)

Description: Python Interactive Signal & Noise Generator

This program implements a GUI with four distinct interactive buttons using Pygame.

Users can switch between White Noise, Pink Noise, Brown Noise, and a Sine Wave in real-time. Mouse coordinates dynamically control panning (X-axis) and volume/amplitude (Y-axis) for the active signal buffer stream.

"""

Members:

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Section: BSCPE Ladderize

```
import ddf.minim.*;
```

```
import ddf.minim.ugens.*;
```

```
Minim minim;
```

```
AudioOutput out;
```



```
// Signals
Oscil sineOsc;
Noise whiteNoise;
Noise pinkNoise;
Noise brownNoise;

// State control
String mode = "SINE";

void setup() {
  size(500, 300);

  minim = new Minim(this);
  out = minim.getLineOut();

  // --- SINE WAVE ---
  sineOsc = new Oscil(440, 0.5f, Waves.SINE);

  // --- NOISE TYPES ---
  whiteNoise = new Noise(Noise.Tint.WHITE);
  pinkNoise = new Noise(Noise.Tint.PINK);
  brownNoise = new Noise(Noise.Tint.BROWN);

  // Start with sine
  sineOsc.patch(out);
}

void draw() {
  background(20);

  fill(255);
  textSize(18);
  text("Audio Signal Generator", 20, 30);

  textSize(14);
  text("Current Mode: " + mode, 20, 60);

  // --- Buttons ---
  drawButton("SINE", 20, 100);
  drawButton("WHITE NOISE", 20, 150);
  drawButton("PINK NOISE", 200, 100);
  drawButton("BROWN NOISE", 200, 150);
```

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```

    text("Click buttons to switch signals", 20, 250);
}

```

```

void drawButton(String label, int x, int y) {
    fill(60);
    rect(x, y, 160, 40, 10);

    fill(255);
    text(label, x + 10, y + 25);
}

```

```

void mousePressed() {

```

```

    // SINE
    if (mouseOver(20, 100)) {
        switchToSine();
    }

```

```

    // WHITE
    if (mouseOver(20, 150)) {
        switchToWhite();
    }

```

```

    // PINK
    if (mouseOver(200, 100)) {
        switchToPink();
    }

```

```

    // BROWN
    if (mouseOver(200, 150)) {
        switchToBrown();
    }
}

```

```

boolean mouseOver(int x, int y) {
    return mouseX > x && mouseX < x + 160 &&
        mouseY > y && mouseY < y + 40;
}

```

```

// ----- SWITCH FUNCTIONS -----

```

```

void stopAll() {
    sineOsc.unpatch(out);
}

```



```
    whiteNoise.unpatch(out);  
    pinkNoise.unpatch(out);  
    brownNoise.unpatch(out);  
}
```

```
void switchToSine() {  
    stopAll();  
    sineOsc.patch(out);  
    mode = "SINE WAVE";  
}
```

```
void switchToWhite() {  
    stopAll();  
    whiteNoise.patch(out);  
    mode = "WHITE NOISE";  
}
```

```
void switchToPink() {  
    stopAll();  
    pinkNoise.patch(out);  
    mode = "PINK NOISE";  
}
```

```
void switchToBrown() {  
    stopAll();  
    brownNoise.patch(out);  
    mode = "BROWN NOISE";  
}
```

IV – Program Output

Conclusion

In this activity, I learned how different audio signals can be generated and modified in real time using Processing and the Minim library. I was able to create a sine wave and compare it with different types of noise, including white noise, pink noise, and brown noise. By using the interactive buttons, I could easily switch between signals and observe their differences. This activity helped me understand the concepts of periodic and non-periodic signals, as well as the relationship between frequency, sound characteristics, and signal generation. Overall, the activity gave me a better understanding of digital audio processing and how signals can be manipulated through programming.

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